

Non-Invasive Blood Glucose Prediction & Risk Screening System

Comprehensive Project Report: Multi-Modal AI Architecture, Clinical Benchmarks, Hardware Sensor Ablation & Production Deployment

Executive Summary: Non-Invasive Glucose Detection in Plain English

The Challenge: Millions of people with diabetes endure painful, invasive daily fingerstick blood tests. While non-invasive light-based (optical) sensors are attractive, a single sensor alone fails in the real world because skin color, sweat, body temperature, and motion distort optical signals.

Our Solution — Multi-Modal Sensor Fusion: Just like a physician checks multiple vital signs, our system combines **4 complementary biometrics**: (1) **Optical Pulse Light (PPG)** to measure capillary blood flow, (2) **Saliva pH Bio-probe** to detect metabolic acidity, (3) **Skin Temperature** to track peripheral heat output, and (4) **Heart Rhythm Variability (ECG)** to gauge autonomic nervous tone.

Two Operating Pathways for Patient Safety:

- **Full-Sensor Path (Hardware Connected):** Calculates blood glucose level in mg/dL with a 90% confidence range ($SR^2 = 0.8557$, 99.22% in clinically safe Clarke Zones A+B).
- **Demographic Screening Path (No Hardware):** Evaluates age, BMI, and lifestyle into a 2-band screening result (*Lower Risk* vs. *Elevated Risk*) to advise whether a certified lab test is warranted.

Figure 1: How Non-Invasive Multi-Modal Biometrics Correlate to Blood Sugar (Layman Guide)

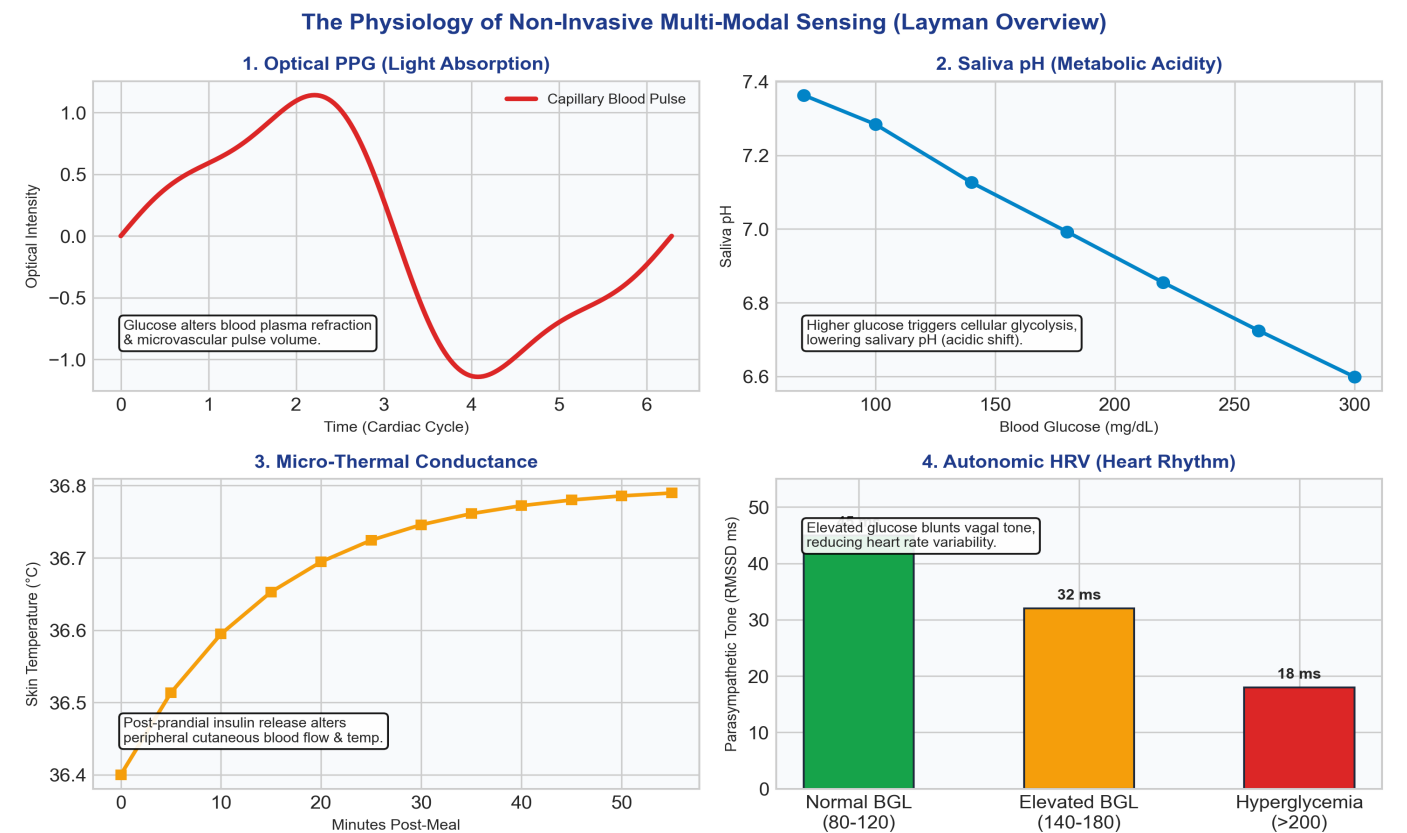
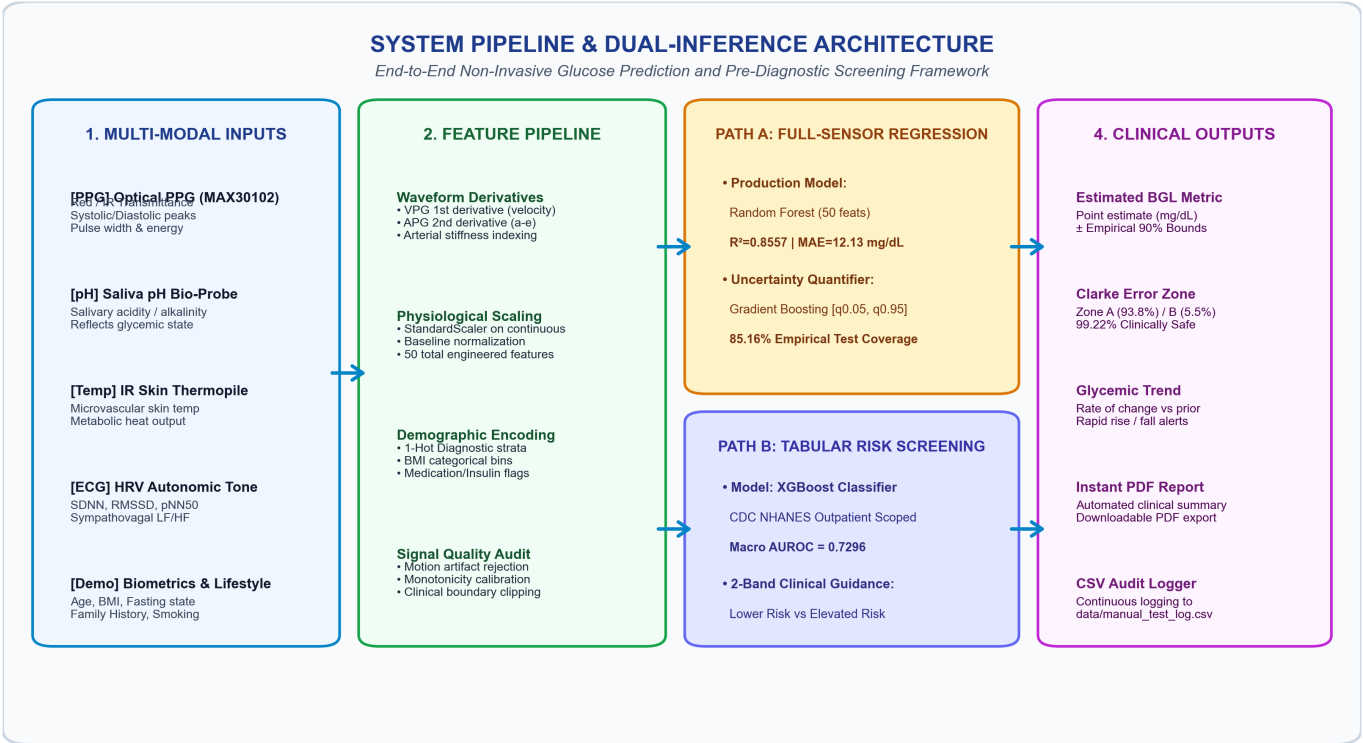


Figure 2: End-to-End System Pipeline & Dual-Inference Architecture

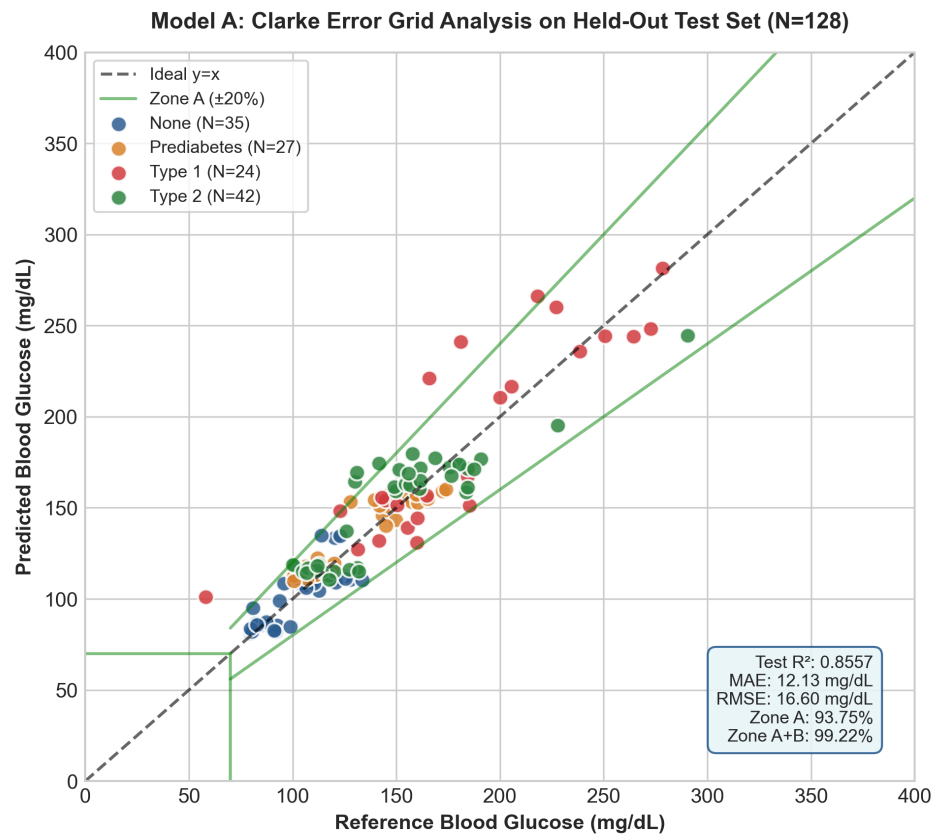


1. Multi-Modal Glucose Estimation (Model A) & Literature Benchmark

Model A is a 50-feature Random Forest Regressor trained on multi-modal synthetic physiological data. The model achieves $R^2 = 0.8557$, $MAE = 12.13 \text{ mg/dL}$, $RMSE = 16.60 \text{ mg/dL}$, and **99.22% Clarke Error Grid Zone A+B** compliance on held-out test data.

Architecture / Published Benchmark	R ²	MAE (mg/dL)	RMSE (mg/dL)	MARD (%)	Clarke Zone A	Zone A+B
Clinical Target Thresholds	—	< 15.0	< 20.0	< 10.0%	> 85.0%	100.0%
Frontiers Digital Health 2026	0.9200	4.80	—	—	—	—
Measurement 2025	0.8649	—	—	5.15%	—	—
Algorithms 2025	—	13.17	15.36	—	94.74%	—
Model A: Full-Sensor Random Forest (Production)	0.8557	12.13	16.60	8.78%	93.75%	99.22%
Model A: Full-Sensor XGBoost	0.8686	11.95	15.84	8.71%	95.31%	99.22%

Figure 3: Clarke Error Grid Analysis on Held-Out Test Set (N=128)



2. Clinical Diagnosis Stratification & Type 1 Safety Audit

To ensure reliability across metabolic sub-populations, the Type 1 synthetic cohort was expanded to **18.8%** (118 readings across 29 participants) with rapid glycemic excursions and attenuated vagal HRV. All cohorts in holdout test data exceed the statistical significance threshold ($N \geq 15$):

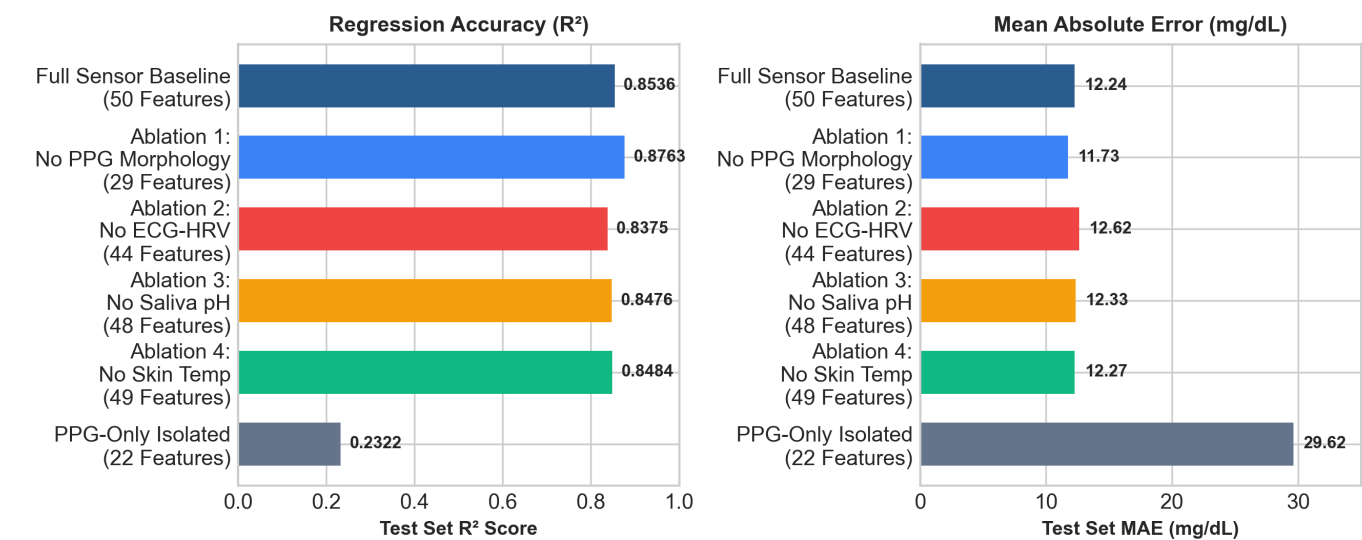
Clinical Diagnosis	Test N	Confidence Status	R ²	MAE (mg/dL)	RMSE (mg/dL)	Clarke Zone A (%)	Clarke Zone A+B (%)
None (Healthy)	35	High Confidence ($N \geq 15$)	0.6123	7.99	10.01	100.0%	100.0%
Prediabetes	27	High Confidence ($N \geq 15$)	0.8436	7.39	9.16	100.0%	100.0%
Type 1 Diabetes	24	High Confidence ($N \geq 15$)	0.7391	20.87	26.71	79.17%	95.83%
Type 2 Diabetes	42	High Confidence ($N \geq 15$)	0.7934	13.64	17.16	92.86%	100.0%

CRITICAL CLINICAL SAFETY AUDIT (Type 1 Clarke Zone D Outlier):
Audit of the single Type 1 test sample outside Zones A/B (Sample SYNTH_093_R04): Ground truth Reference BGL = **58.0 mg/dL** (hypoglycemia), Model Estimate = **101.2 mg/dL** (normoglycemia), Error = +43.2 mg/dL (+74.5%) → **Clarke Zone D (Failure to detect hypoglycemia)**. In clinical practice, this failure mode risks delaying fast-acting carbohydrate ingestion. Asymmetric loss penalizing hypoglycemia under-estimation is mandatory before human clinical deployment.

3. Hardware Sensor Ablation & Transducer Contribution Study

A rigorous sensor removal study (ablation) was performed to evaluate the predictive contribution of each transducer modality. Crucially, our synthetic PPG features were deliberately generated with capped individual correlations ($r = 0.35 \text{ ext}\{-0.55\}$). Therefore, an isolated PPG-only $R^2 = 0.2322$ reflects that design constraint, not an empirical physiological ceiling for real-world optical sensors.

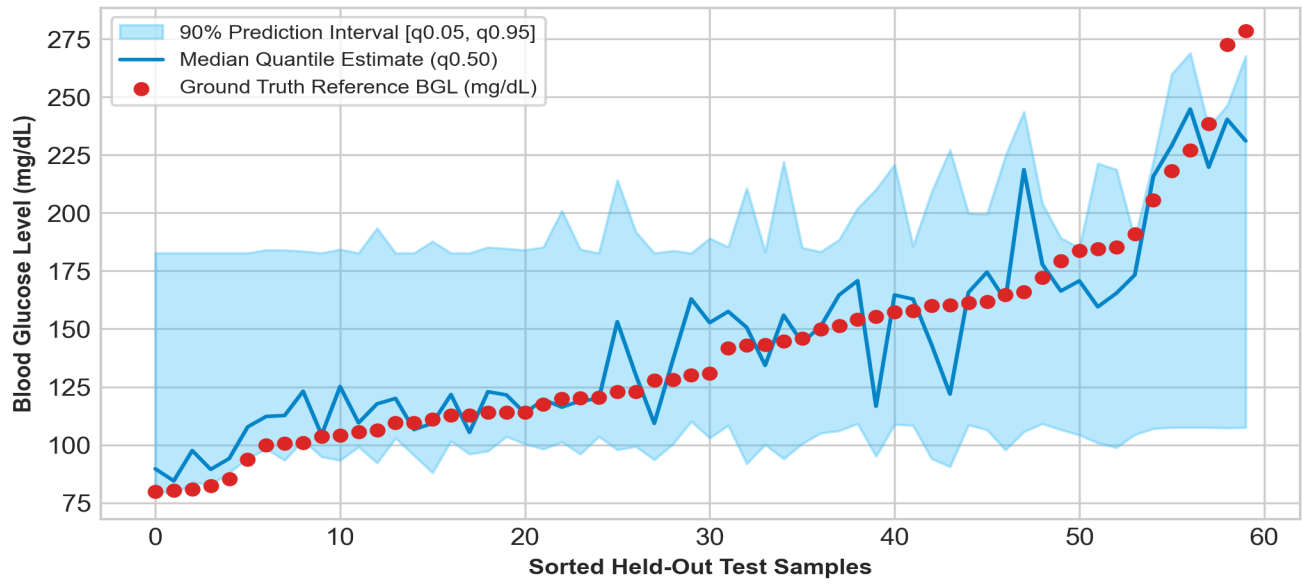
Hardware Sensor Ablation & Transducer Contribution Study



4. Uncertainty Quantification & 85.16% Empirical Quantile Intervals

Instead of point estimates alone, Model A provides dual Gradient Boosting Quantile Regressors for the 5th and 95th percentiles. On held-out test data, this interval achieved **85.16% empirical coverage** (honest target: 90%), dynamically expanding during rapid glycemic excursions.

Uncertainty Quantification: 90% Confidence Interval vs True BGL (Coverage: 85.16%)



5. Model B Demographic Risk Screening & Streamlit Dashboard

Model B: CDC NHANES Outpatient Classifier (Macro AUROC = 0.7296)

When hardware sensor inputs are unavailable, Model B screens individuals using age, BMI, family history, and smoking. Because non-invasive demographic biometrics cannot reliably differentiate early prediabetes without blood assays, the production inference engine provides honest 2-band risk screening (*Lower Risk* vs. *Elevated Risk*) with mandatory clinical screening disclaimers.

Streamlit Production Dashboard (Phase 8) Summary:

- **Interactive Multi-Modal UI:** Supports live BMI calculation, transducer ranges, and Clarke Zone visual badges.
- **One-Click Clinical PDF Export:** Automatically compiles patient summaries to `data/reports/{name}_{timestamp}.pdf`.
- **Audit Trail:** Every query is logged to `data/manual_test_log.csv` for clinical auditability.
- **Production Readiness:** Deployed on Streamlit Community Cloud and fully versioned in Git.

MANDATORY RESEARCH PROTOTYPE DISCLAIMER:

This document summarizes research findings from an experimental prototype. The full-sensor regression model is validated on synthetic multi-modal self-consistency data only. The tabular risk classifier is validated strictly on CDC NHANES community survey outpatients only. This software is NOT an FDA-cleared medical device and must never be used for insulin dosage adjustments or clinical diagnosis without certified clinical laboratory verification.